2	**(a)**	(lift is perpendicular to the wings) the wings are tilted so that the lift has a horizontal component to provide the centripetal force	A1
	(b)	$v = \dfrac{2\pi r}{T} = \dfrac{2\pi \times 12000}{250}$	C1
		301.6 m s^{-1}	A1
	(c)	$L\sin\theta = \dfrac{mv^2}{r}$	C1
		$L\cos\theta = mg$	C1
		first eq / second eq $\Rightarrow$ $\tan\theta = \dfrac{v^2}{rg} = \dfrac{(301.6)^2}{12000 \times 9.81} \Rightarrow \theta = 37.7°$	A1
	(d)	larger θ so L cos θ < mg [B1] $\Rightarrow$ airplane falls [B1] OR L sin θ increases so r decreases [B1] $\Rightarrow$ airplane moves in smaller circle [B1] (airplane moves in a downward helical path of smaller radius)	

3	(a)	A real gas approaches ideal gas behaviour at high temperature and low